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MODEL SELECTION AND VALIDATION REPORT ( boreal_biomass model )
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## OBSERVATIONS:863

## SANITY: warning − sanity check failed for the following models:
spatial_random_fields_with_anisotropy

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1_with_anisotropy

## MODEL SUMMARY ( boreal_biomass model )

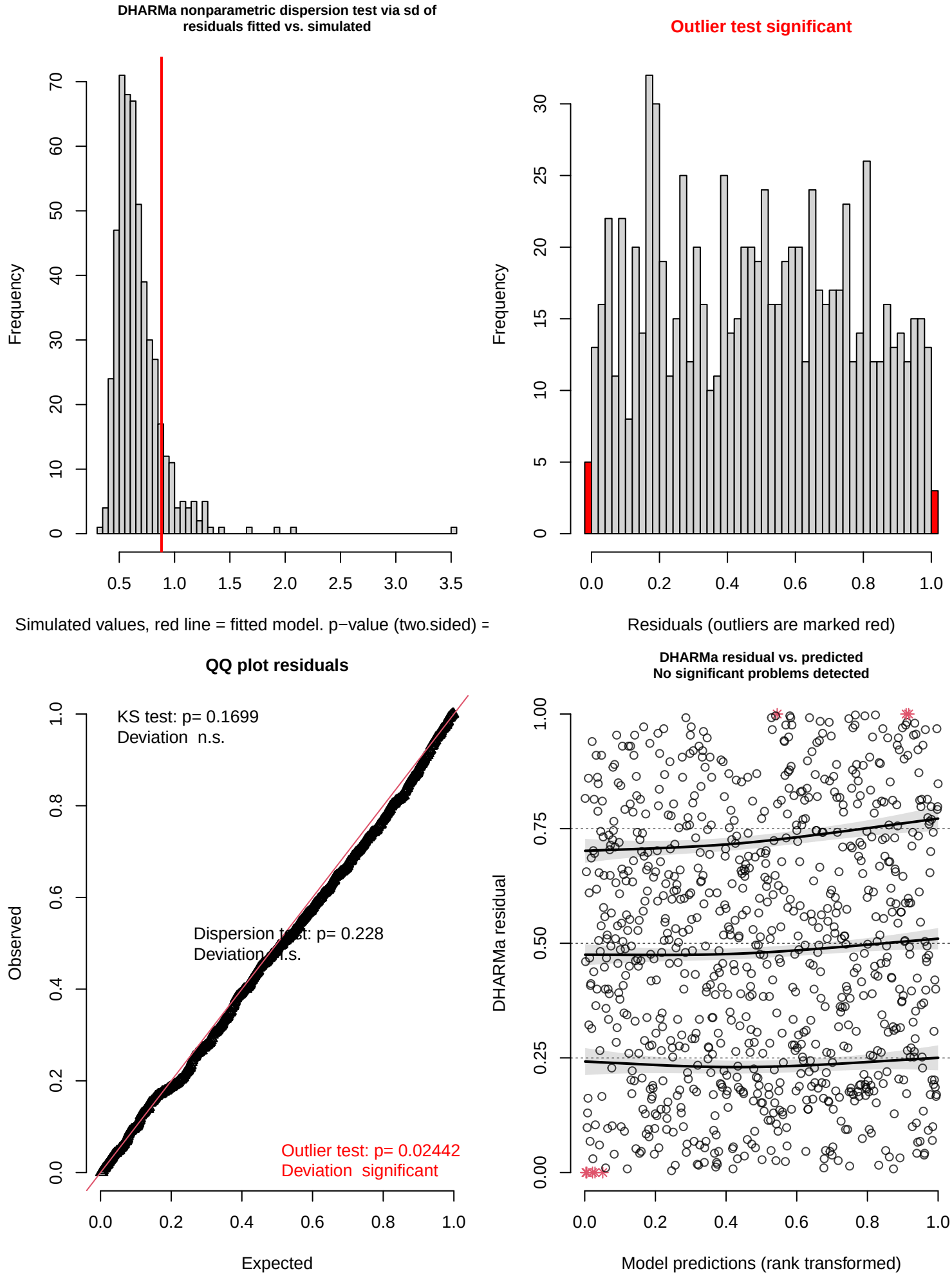
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

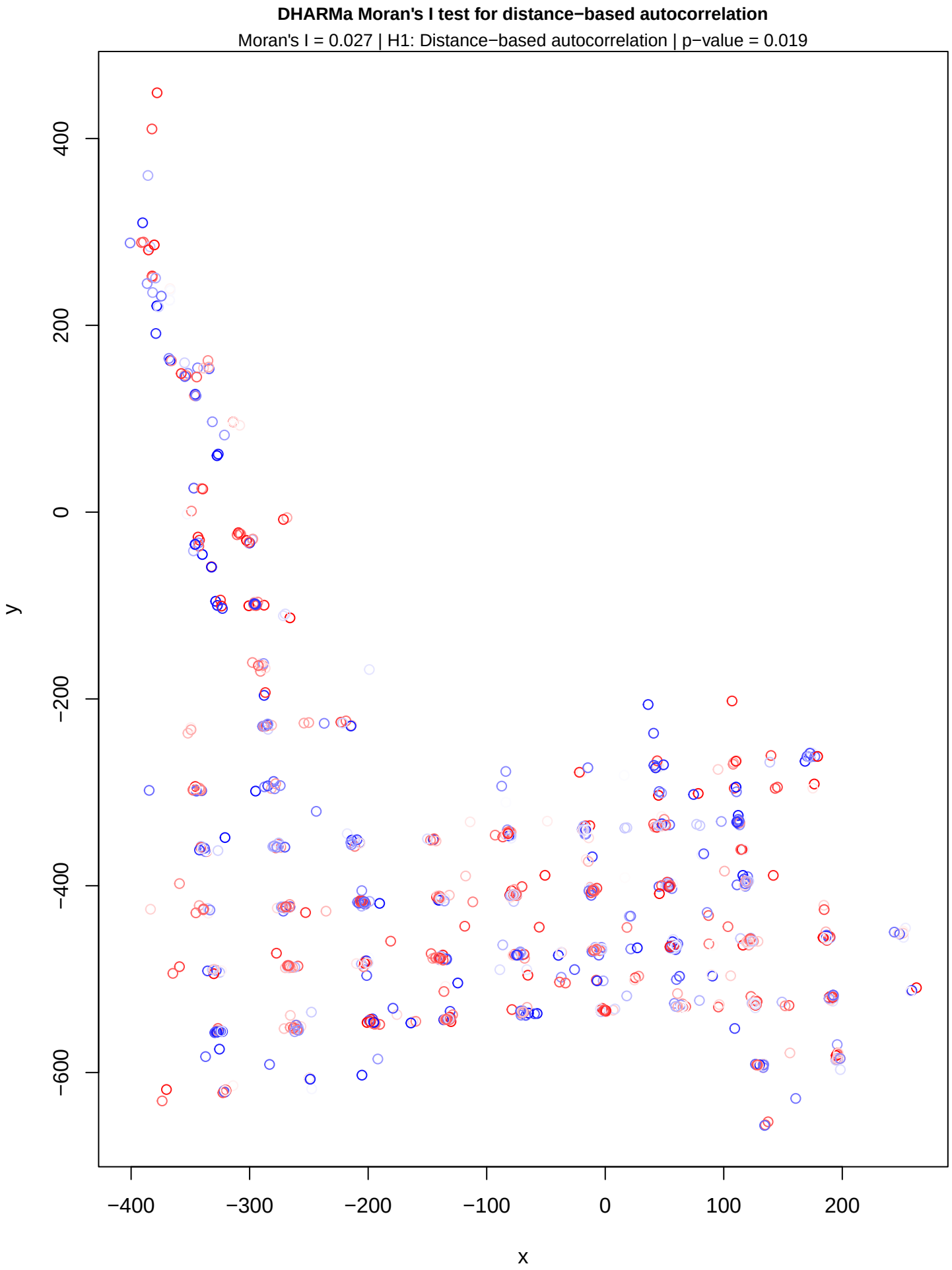
Conditional model:
              coef.est coef.se
(Intercept)       7.46    2.84
kde_fric          0.00    0.04
kde_feve         −0.12    0.03
tric              0.04    0.03
teve             −0.34    0.03
sbt_mean          0.06    0.09
sbt_sd            0.07    0.04
sst_sd           −0.07    0.06
log_chla_mean    −0.03    0.04
depth            −0.23    0.05

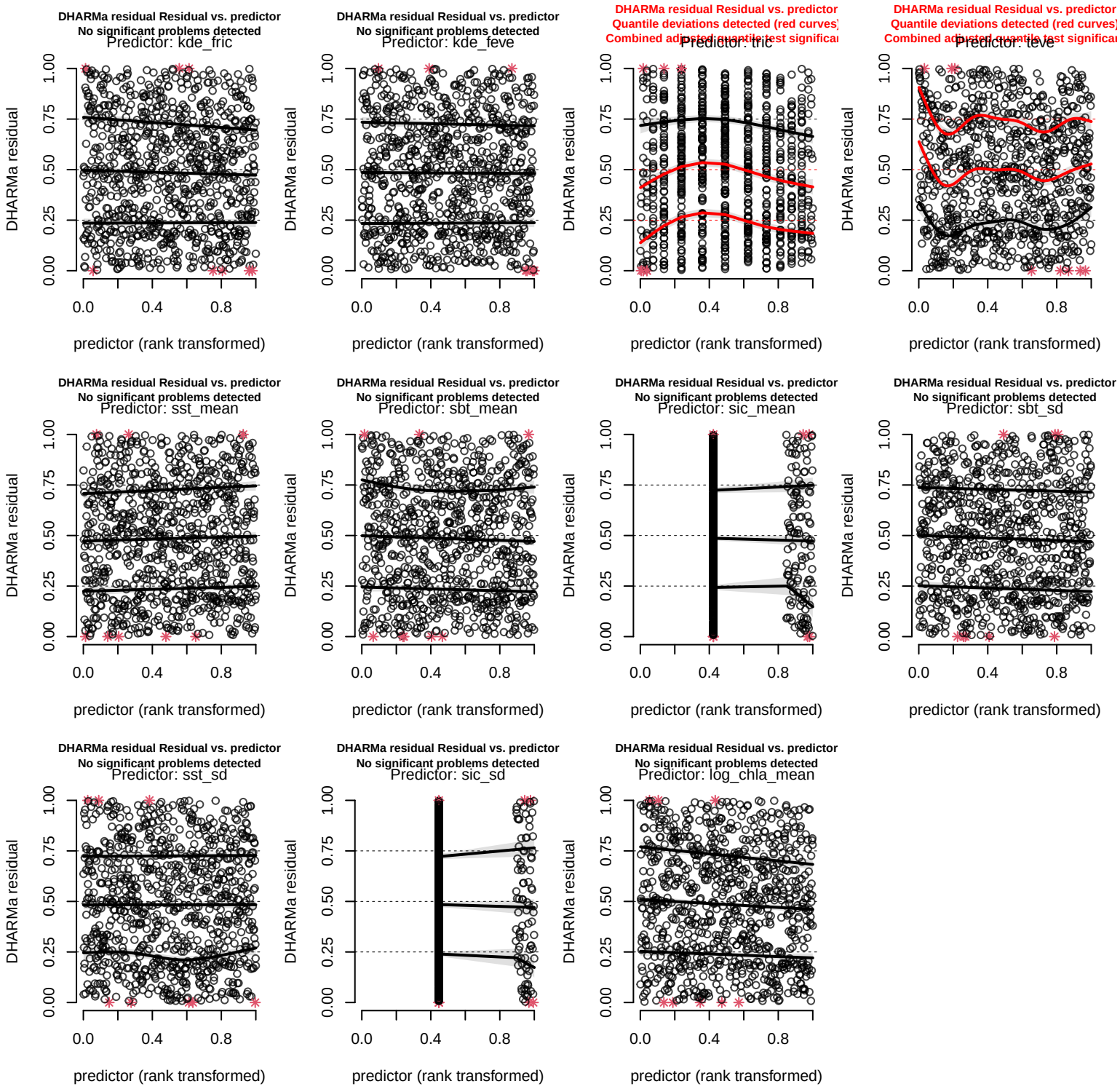
Dispersion parameter: 0.71
Spatiotemporal AR1 correlation (rho): 1.00
Matérn anisotropic range (spatiotemporal): 508.4 to 3330.2 at 96 deg.
Spatiotemporal marginal AR1 SD: 1.64
REML criterion at convergence: 7140.162

See ?tidy.sdmTMB to extract these values as a data frame.
See ?plot_anisotropy to plot the anisotropic range.

## SELECTION STATISTICS
                                                   df      AIC delta_AIC
spatiotemporal_random_fields_ar1_with_anisotropy 16 14312.32  2.617889
spatiotemporal_random_fields_ar1                 14 14314.94 17.170362
spatiotemporal_random_fields_iid_with_anisotropy 15 14332.11  1.560288
spatiotemporal_random_fields_iid                 13 14333.67 24.197603
spatial_random_fields                            13 14357.87  1.464404
spatial_random_fields_with_anisotropy            15 14359.33 26.799480
error_only                                       11 14386.13        NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
Simulated values, red line = fitted model. p−value (two.sided) =
Outlier test significant
Frequency
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.1699
Deviation n.s.
Dispersion test: p= 0.228
Deviation n.s.
Outlier test: p= 0.02442
Deviation significant
Observed
Expected
DHARMa residual vs. predicted
No significant problems detected
DHARMa residual
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.027 | H1: Distance−based autocorrelation | p−value = 0.019
y
400
200
0
−200
−400
−600
x
−400
−300
−200
−100
0
100
200


DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: kde_fric
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: kde_feve
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significan
Predictor: tric
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significan
Predictor: teve
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sst_mean
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sbt_mean
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sic_mean
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sbt_sd
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sst_sd
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sic_sd
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: log_chla_mean
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8
predictor (rank transformed)